AnalogtoKey

User Guide

Map joystick & arcade stick buttons to keyboard keys

Version 0.2 Beta - Windows 10 / 11

Contents

1 What is AnalogtoKey?

AnalogtoKey is a small Windows utility that reads input from joysticks and arcade sticks and converts every button press, D-pad direction, and axis movement into real keyboard keystrokes - exactly as if you had pressed a key on your keyboard.

The problem it solves

Some games - such as Running Train - detect any connected USB controller and try to interpret it as an Xbox gamepad, causing incorrect or doubled input. These games do support keyboard input correctly. AnalogtoKey bridges the gap:

* Hides the controllers from all other applications using the HidHide kernel driver
* Reads all input directly itself
* Sends the correct keystrokes to the active game window

When do you need it?

Running Train / similar sims	Game forces gamepad mode on connected sticks - use AnalogtoKey
HORI Fighting Stick Mini	Detected as Xbox controller - causes wrong axis behaviour
Two sticks at once	Full individual mapping for each controller simultaneously
TDW6 and similar	Not needed - controller input can be disabled in-game

Key features

* Map any button (1-16) to any keyboard key
* Map all 8 D-pad directions independently
* Map analog axes (X/Y/Z/Rx/Ry/Rz) with adjustable step count
* Multiple named profiles - switch instantly between games
* Runs in the system tray - always available, never in the way
* Automatic HidHide management - hides on start, restores on exit

2 Installation

Requirements

OS	Windows 10 or Windows 11 (64-bit)
.NET runtime	None - the installer is self-contained
HidHide driver	Required for device hiding (bundled in installer)
USB ports	One per controller

Installer walkthrough

1. Run AnalogtoKey_Setup.exe - right-click and Run as administrator if prompted.
2. Choose components - the installer offers to install HidHide (the device-hiding driver) as an optional component. Strongly recommended for use with Running Train.
3. Complete the installation - files are copied to C:\Program Files\AnalogtoKey\.
4. Restart Windows if you installed HidHide - the kernel driver requires a reboot.
5. Launch AnalogtoKey from the Start Menu or desktop shortcut.

About HidHide

HidHide is a free, open-source Windows kernel driver that hides USB HID devices from selected applications. AnalogtoKey uses it to make your controllers invisible to every other program while AnalogtoKey itself continues to read them.

When AnalogtoKey starts, it automatically adds itself to the HidHide whitelist and hides all detected controllers. When AnalogtoKey exits, it removes itself - controllers become visible again immediately.

HidHide not installed?

If HidHide is missing, AnalogtoKey shows a yellow warning banner with an Install button. Click it and follow the prompts. A restart is required after installation.

Uninstall

Use Windows Settings > Apps > AnalogtoKey > Uninstall. Profiles are stored separately in %AppData%\AnalogtoKey\profiles\ and are not removed automatically - delete that folder manually if desired.

3 The Interface

The main window is divided into six areas. The screenshot below shows a fully configured profile with one controller connected.

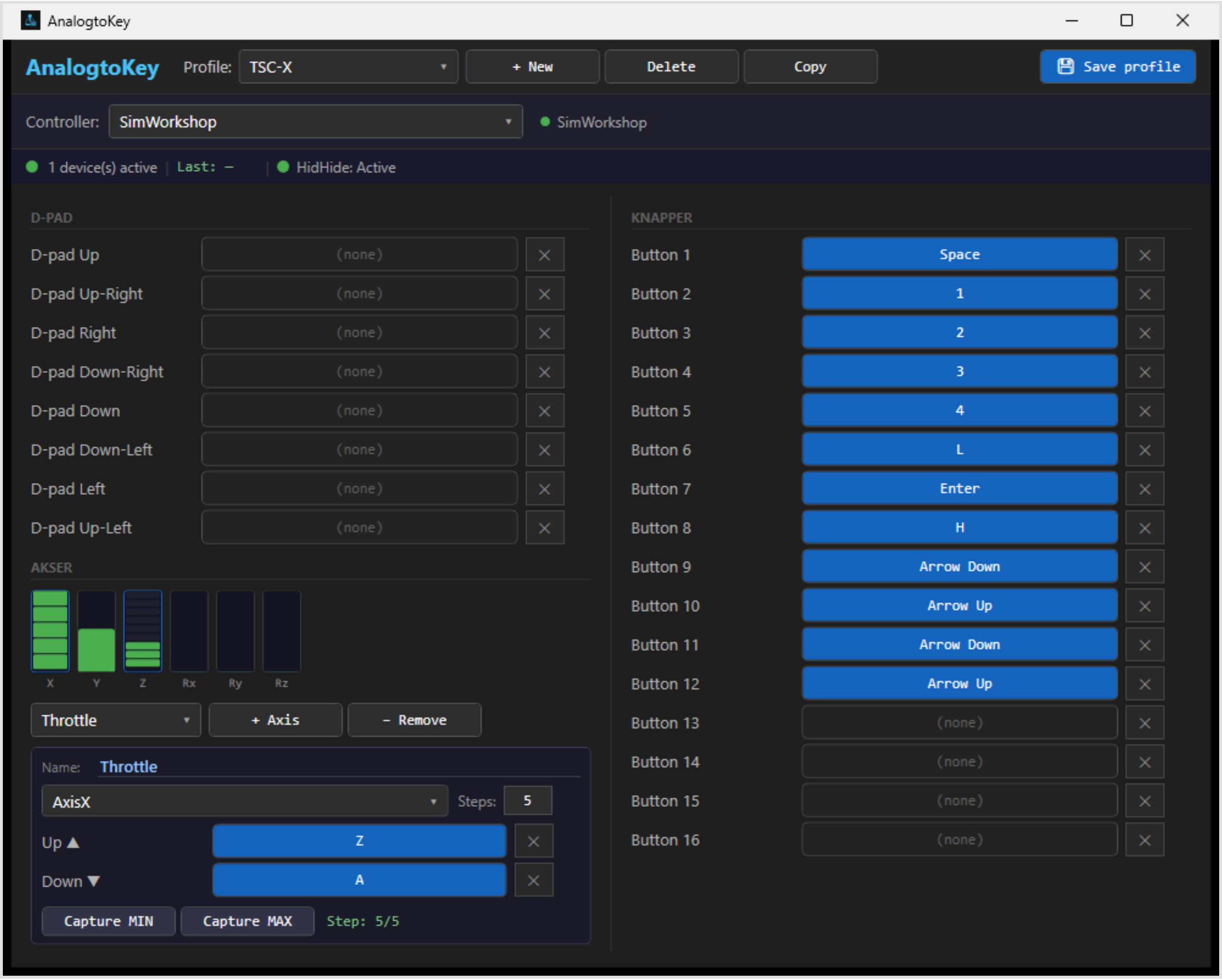

Figure 1 - AnalogtoKey main window with a profile loaded

Area overview

(1) Profile bar	Drop-down showing the active profile name. Buttons: + New (create blank profile), Delete (remove current), Copy (duplicate), Save Profile (write to disk). Unsaved changes are lost if you switch profiles without saving.
(2) Controller selector	Drop-down listing all connected controllers. The green dot confirms the device is active. Only the selected controller's mappings are shown below.
(3) Status bar	Three live indicators: number of active devices, the last key sent, and HidHide status (Active / Inactive).

(4) D-PAD section	Eight direction slots - Up, Up-Right, Right, Down-Right, Down, Down-Left, Left, Up-Left. Click any slot then press a key to assign. Click x to clear.
(5) AKSER (Axes) section	Six vertical bars visualise live axis position (X/Y/Z/Rx/Ry/Rz). The panel below the bars shows the selected axis detail (see Chapter 5).
(6) KNAPPER (Buttons) section	16 button slots. Blue = assigned, dark = unassigned. Click any slot then press a key to assign.

4 Setting up a Profile

A profile stores a complete set of mappings - D-pad, buttons, and axes - for all your connected controllers. You can have as many profiles as you like and switch between them instantly.

Create a new profile

1. Click + New in the profile bar. A new blank profile appears.
2. The profile name field becomes editable. Type a name (e.g. Running Train) and press Enter.
3. Select your controller in the Controller drop-down.
4. Assign keys to buttons and D-pad directions (see below).
5. Click Save Profile to write the profile to disk.

Assigning a key to a button

1. Locate the button slot in the KNAPPER section (Button 1 to Button 16).
2. Click the slot - it turns into a blinking capture field.
3. Press the key you want on your keyboard (e.g. Space, W, F1).
4. The slot shows the captured key name and turns blue.
5. Repeat for each button you want to map.

Assigning a key to a D-pad direction

The process is identical to buttons. In the D-PAD section, click any direction slot (e.g. D-pad Up), then press the desired key. All eight directions can be mapped independently - or left as (none) to be ignored.

Clearing a mapping

Click the x button next to any slot to remove its assignment. The slot returns to (none) and that input will be ignored.

Remember to save

Profile changes exist only in memory until you click Save Profile. Switching profiles or closing the app will discard unsaved changes.

Switching profiles

Open the profile drop-down and select any saved profile. The mappings load immediately and take effect for new input - no restart required.

5 Setting up Axes

Axes handle analog inputs - throttle levers, joystick tilt, rotary knobs, and pedals. Each physical axis (X, Y, Z, Rx, Ry, Rz) is divided into a number of steps. As the axis moves through each step boundary, AnalogtoKey sends the configured key down or up.

Add an axis mapping

1. In the AKSER section, click + Axis. A new entry appears in the selector drop-down.
2. Give it a descriptive name in the Name field (e.g. Throttle, Brake).
3. Choose the physical axis: AxisX, AxisY, AxisZ, AxisRx, AxisRy, or AxisRz.
4. Set the number of Steps (1-8). Steps = 1 means binary on/off. Steps = 5 means five evenly-spaced zones - useful for a throttle lever.
5. Click the Up (triangle up) slot and press the key for the positive direction.
6. Click the Down (triangle down) slot and press the key for the negative direction.
7. Calibrate the range - see Calibrating MIN / MAX below.

Calibrating MIN / MAX

Raw axis values from DirectInput vary between controllers. Calibration tells AnalogtoKey what value corresponds to the physical minimum and maximum, so step thresholds are calculated correctly.

1. Move the axis to its minimum position (fully back, fully released). Click Capture MIN.
2. Move the axis to its maximum position (fully forward, fully pressed). Click Capture MAX.
3. The Step: N/N indicator shows which step the axis is in. Move the axis to verify.

Step indicator

The Step: N/N label (e.g. Step: 3/5) shows the current step out of total configured. Use it to confirm the axis is being read correctly after calibration.

How stepped axes work - example with Steps = 5

With Steps = 5, Up key = Z, Down key = A:

Steps 4-5 (top zone)	Z held down continuously
Steps 2-3 (middle zone)	Neither key - neutral
Step 1 (bottom zone)	A held down continuously

Remove an axis mapping

Select the axis in the drop-down and click - Remove. The entry is deleted from the profile. Save the profile afterwards.

6 Tips & Tricks

Copy a profile for a new game

Open the profile you want to base the new one on, then click Copy. A duplicate is created with a new name. Rename it and adjust the mappings. The original is unchanged.

Use descriptive axis names

Rename axes to match what they control (Throttle, Brake, Rudder). The name appears in the drop-down, making it easy to select the right axis at a glance.

Two controllers simultaneously

Both controllers appear in the Controller drop-down. Switch between them to edit each one's mappings within the same profile. Both are active at the same time during use - you only need to switch the selector when editing.

Minimise to system tray

Clicking the X button hides the window to the system tray - it does NOT quit the app. AnalogtoKey continues running and forwarding keystrokes in the background. Double-click the tray icon to restore the window. Right-click for Open / Exit.

Profiles auto-load on start

AnalogtoKey remembers the last profile you used and loads it automatically at startup. You rarely need to touch the profile selector once it is set up.

HidHide is optional for some games

If your game lets you disable gamepad input in its own settings, HidHide is not needed. AnalogtoKey still works - it just cannot prevent the game from also seeing raw controller data.

Profile files are plain JSON

Profiles are stored in %AppData%\AnalogtoKey\profiles\ as human-readable JSON files. Back them up, share them, or copy to another PC by copying the .json files.

Reverse an axis direction

AnalogtoKey always maps positive axis movement to the Up key and negative movement to the Down key. If your physical axis is wired the wrong way round (throttle forward fires the wrong key), simply swap the two key assignments: put the key you want for forward in the Down slot and the key for backward in the Up slot. The axis will now behave as expected.

Shorten the active travel zone

You do not have to calibrate the full physical range. To pack all steps into the first half of the lever travel, move the lever to the bottom, click Capture MIN, then move it only halfway and click Capture MAX. All steps now fit in that shorter range - pushing past the midpoint has no further effect (the axis stays clamped at the top step). Conversely, set Capture MIN slightly above the resting position to create a dead zone at the start so accidental bumps do not fire a key.

AnalogtoKey v0.2 Beta - Profiles: %AppData%\AnalogtoKey\profiles